Bayesian Inference of Athletic Performance

Ellie Wakefield (79818035)
Aryan Ballani (61663514)

April 2026

1 Contribution Statement

Ellie focused on the problem formulation, literature review, and the primary technical writing and visualization. Aryan handled the Strava API data collection pipeline, the implementation of the `simPPLe` importance sampling framework. Both members collaborated on the Bayesian model design, the development of the Stan state-space model, and posterior diagnostics.

2 Introduction

We are interested in analysing Ellie's underlying running fitness over time based on observed data collected through Strava, a workout-tracking app. We take a Bayesian approach, utilising a naïve model and a more sophisticated model, and multiple implementation approaches.

2.1 Research Question

Based on observed training data (weekly mileage, pace, and elevation), how can we use a Bayesian state-space framework to estimate Ellie's latent fitness level, X_T, for her marathon race week?

2.2 Literature review

There has been some research in the area of Fitness-Fatigue modelling within a Bayesian framework. Marchal et al. (2025) implemented a Fitness-Fatigue Model using a Bayesian model with biologically-motivated prior choices but encountered issues including poor identifiability of fitness and fatigue parameters. Moxnes and Hausken (2008) used a mathematical dynamic model to quantify an optimal rest period to maximise performance, and related results. To our knowledge, there has not been any work in applying a Bayesian model to Strava data with the goal of predicting marathon fitness.

3 Exploratory Data Analysis (EDA)

Before modeling, we performed EDA to characterise key features of the data and justify our modeling choices.

3.1 Data Summary

The dataset consists of 319 weeks of activity tracking. Table 1 summarizes variables of interest, namely weekly mileage, average pace, and elevation gain.

Table 1: Summary Statistics for Strava Training Data

Variable	N	Mean	Std. Dev	Min	Median	Max
Weekly Mileage (km)	319	23.40	18.70	0.00	21.10	70.30
Avg. Pace (min/km)	270	5.37	0.51	4.63	5.27	8.89
Elevation Gain (m)	319	194.00	193.00	0.00	140.00	1447.00

Statistical Insights: The primary observation variable is `weekly_mileage`. We treat the "Marathon Week" as the final time step T in our sequence. Several key trends emerged from this summary:

- **High Volatility:** The standard deviation for mileage (18.70 km) is nearly as large as the mean (23.40 km), confirming the presence of significant training "peaks" and "troughs" (injury or rest weeks) mentioned in the visual analysis.
- **Pace Consistency:** Despite the variance in volume, the average pace is relatively stable with a median of 5.27 min/km. The discrepancy in N for pace (270 vs 319) indicates weeks with zero training where pace was undefined, which we handled by imputing a baseline for the state-space model.

3.2 Visualizations

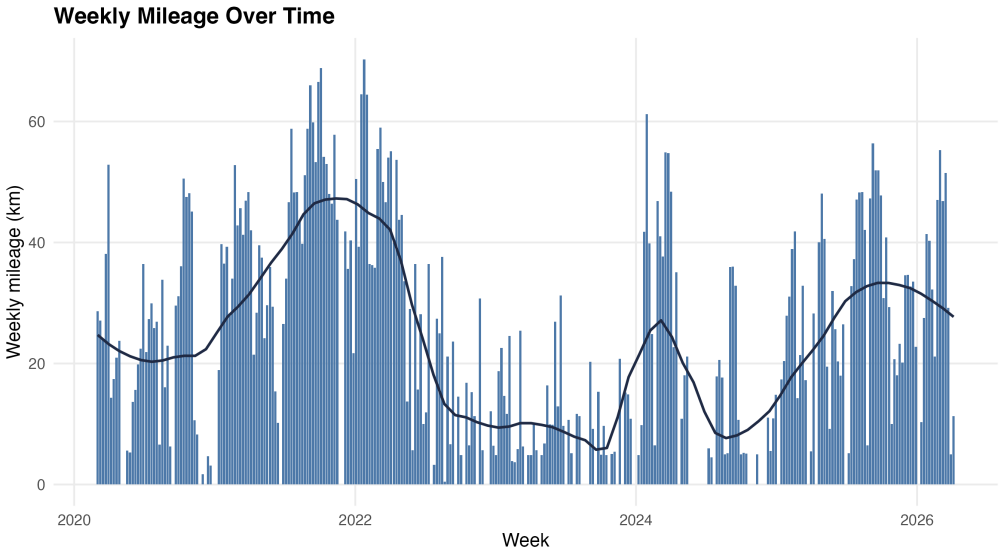


Figure 1: Weekly Mileage Over Time (km)

The training data exhibits significant volatility, with several high-volume peaks exceeding 50 km followed by sharp drops to near-zero mileage. The increase in mileage around mid-2024, noticeable in Figure 1, roughly marks the beginning of intentional marathon training.

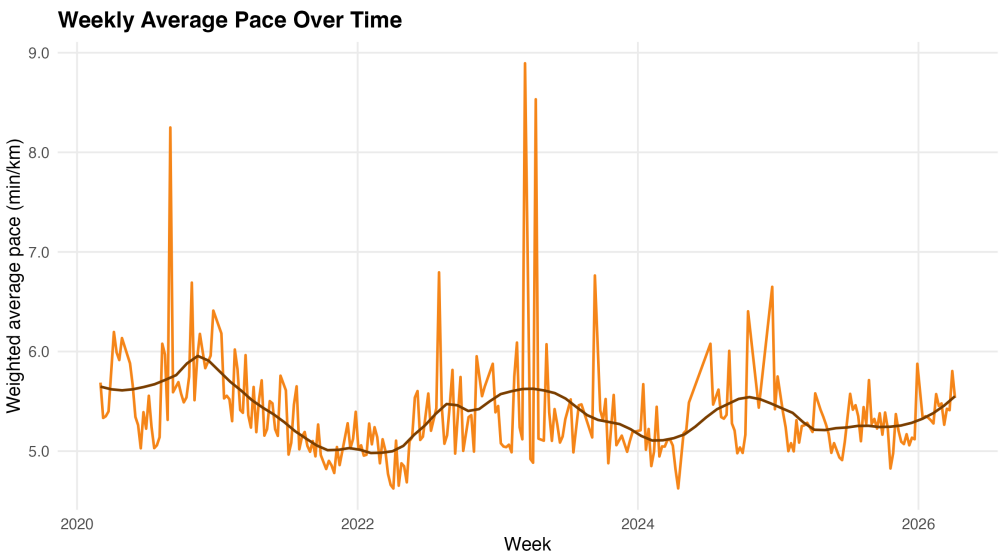


Figure 2: Weekly Average Pace (min/km) Over Time

We notice that pace also exhibits a similar pattern of volatility, but in an inverse fashion. The pace seems to drop around 2022 followed by a sudden and sharp increase, indicating an injury/recovery period, followed by a drop. Post 2024, the pace is more or less consistent, which points towards the intentional marathon training period as well.

Comparing trends from Figures 1 and 2 suggests an inverse relationship between volume and pace, which is interesting to see because as Ellie starts her training, she is significantly pushing mileage while at the same time maintaining a good pace.

4 Methodology

4.1 Model 1: Naïve model (simPPLe implementation)

The naïve model serves as our baseline, utilizing Self-Normalized Importance Sampling (SNIS) to approximate the posterior. In this version, we treat each week's training as an independent realization of a latent fitness variable.

For each week $i \in \{1, \ldots, n\}$, we define the model as follows:

$$\begin{aligned} X_i &\sim \text{Uniform}(0,1) && \text{(Prior Fitness Level)} \\ Y_i|X_i &\sim \text{Gamma}(\alpha = 50X_i, \theta = 1) && \text{(Likelihood of Weekly Mileage)}, \end{aligned}$$

where α is a shape parameter and θ is a scale parameter, that is, $\mathbb{E}[Y_i|X_i] = \alpha \cdot \theta = 50X_i$. This definition allows us to interpret the expectation of the likelihood as linearly related to underlying fitness. For example, a maximum fitness value of $X_i = 1$ at week i signifies, under the likelihood, an ability to run 50km in week i.

We justify this particular specification of the likelihood by the assumption that Ellie defines her training week-to-week by determining her mileage based on how she's feeling. For example, if her legs are in great shape, i.e. $X = 1$, she will decide to run 50km this week. If her legs are tired or she's trying to avoid injury, i.e. $X = 0.5$, she will decide to run 25km. If she has a stress fracture or develops a serious flu, i.e. $X = 0$, she will skip training this week and run 0km.

We use `simPPLe` for a simple implementation of this naïve model, we employ the framework, with a modification to accumulate weights in log-space to prevent numerical underflow:

$$\log(W) = \sum_{i=1}^{n} \log\left[p(y_i|x_i)\right]$$

4.2 Model 2: Sequential model (Stan implementation)

To build upon our initial naïve model to a more sophisticated model, we introduce a sequential structure which allows latent fitness at a given week i to depend on fitness during the previous week $i-1$. This is a more realistic approach, as it reflects the real-world truth that fitness evolves over time. We define the model as follows:

$$\begin{aligned} X_i|X_{i-1} &\sim \text{logistic}\left(N(X_{i-1}, 1)\right) \\ Y_i|X_i &\sim \text{Gamma}(\alpha = 50X_i, \theta = 1). \end{aligned}$$

We chose to use a Normal distribution for our hierarchically-structured prior as it allows X_i to be centered at X_{i-1}, with σ chosen to be 1. We then employ a logistic transformation to ensure the fitness variable remains in the support $(0, 1)$.

We implemented this model in Stan (code shown in 6.2). We include a `transformed parameters` block for the logistic transformation in the prior, and a `generated quantities` block to predict the fitness level at week "T", that is, X_T.

5 Results and Interpretation

5.1 Model 1 Results

Using the `simPPLe` importance sampling framework with 10,000 iterations, the posterior mean for Ellie's marathon-week fitness was found to be:

$$\hat{X}_T \approx 0.4973.$$

Interpreting this through our likelihood ($\alpha = 50X_T$), this suggests an ability to run approximately **24.87 km** for the race week. While this is lower than a full marathon distance, it reflects the global mean of the training history, which includes non-active weeks. This is a naive approach and more so highlights the necessity of the Model 2 State-Space approach to capture recent momentum, keeping in mind the influence of the previous week's fitness level on the current week's.

5.2 Model 2 Results

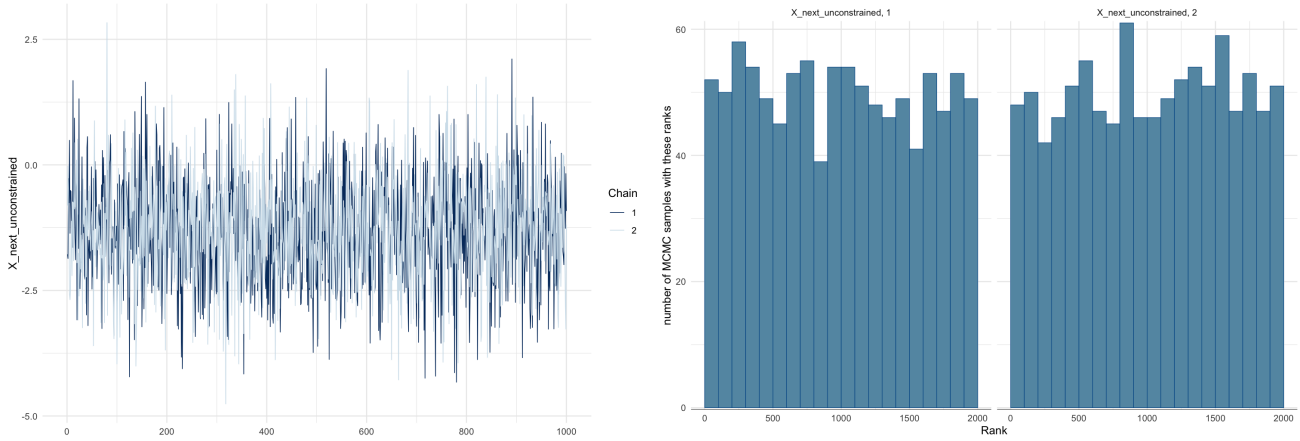
The sophisticated, sequential model implemented in Stan predicts a value of:

$$\hat{X}_T \approx 0.255$$

with an 80% credible interval given by $[0.065, 0.510]$. While a 25% fitness level is not what Ellie may hope for at this point in her training, the result makes sense given mileage in recent weeks (Ellie has been having knee pain and reducing mileage). The large width of the credible interval also indicates large uncertainty in this result.

5.3 Model diagnostics

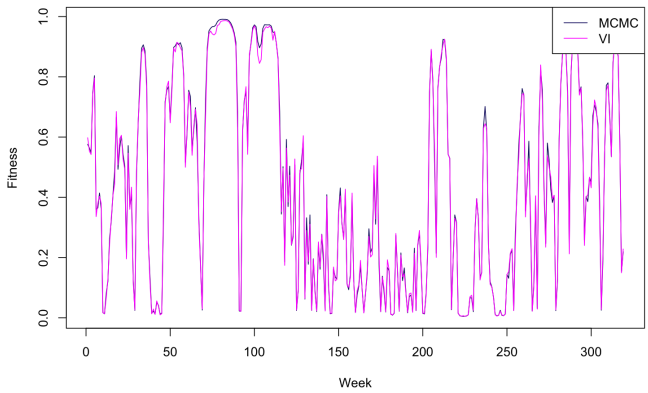
We produced a trace plot and rank plot for our Stan-implemented model, displayed below.



Both plots indicate fast mixing between the two chains (no discernible pattern in the trace plot; rank histograms approximately uniform).

We also conducted an invariance test that resulted in a p-value of $p = 0.341$, which does not suggest the presence of a bug.

Finally, we fitted a variational inference approach and compared its posterior approximation with that from MCMC, shown below.



The posterior approximations are very similar, suggesting agreement between the two methods and valditating our model implementation.

6 Discussion

6.1 Assumptions and Dataset Decisions

We assumed that the relationship between fitness and mileage capacity is linear. We chose to focus on the latter portion of the dataset for the naïve model to ensure the "stationary" assumption wasn't violated by the drastic increase in marathon-specific training.

6.2 Limitations

The model has an extremely over-simplified structure, in the interest of interpretability and time constraints in implementation, which does not reflect intricacies in real-world training. For example, it is common practice to "taper" mileage leading up to race day to give one's legs time to recover while maintaining fitness gained from higher-mileage training blocks. This is not reflected in the model, which assumes a positive linear trend between weekly mileage and fitness. The model also does not incorporate factors such as speed training, elevation gain, recovery practices and nutrition which also have a significant impact on running fitness.

A Appendix

Full updated code is available on the GitHub repo.

Model 1: simPPLe Implementation

```
# Modified SNIS logic with Log-Sum-Exp for numerical stability
log_weight <<- 0.0

observe = function(realization, distribution) {
  log_weight <<- log_weight + d(distribution)(realization, log = TRUE)
}

posterior = function(ppl_function, number_of_iterations) {
  samples <- numeric(number_of_iterations)
  log_weights <- numeric(number_of_iterations)
  for (i in 1:number_of_iterations) {
    log_weight <<- 0.0
    samples[i] <- ppl_function()
    log_weights[i] <- log_weight
  }
  max_log_w <- max(log_weights)
  weights <- exp(log_weights - max_log_w)
  return(sum(samples * weights) / sum(weights))
}

fitness_function = function() {
  for (i in seq_along(weekly_mileage_obs)) {
    fitness = simulate(X)
    obs = max(weekly_mileage_obs[i], 0.01)
    observe(obs, Gammad(scale = 1, shape = 50*fitness))
  }
  return(simulate(X))
}
```

Model 2: Stan State-Space Implementation

```
data {
  int<lower=0> N;
  vector[N] y;
}
parameters {
  vector[N] X_unconstrained;
}
transformed parameters {
  vector<lower=0, upper=1>[N] X = inv_logit(X_unconstrained);
}
model {
  X_unconstrained[1] ~ normal(0,1);
  y[1] ~ gamma(50 * X[1], 1);
  for (i in 2:N) {
    X_unconstrained[i] ~ normal(X_unconstrained[i-1], 1);
    y[i] ~ gamma(50 * X[i], 1);
  }
}
generated quantities {
  real X_next_unconstrained = normal_rng(X_unconstrained[N], 1);
  real X_next = inv_logit(X_next_unconstrained);
}
```

References

Marchal, A., Benazieb, O., Weldegebriel, Y., Méline, T., and Imbach, F. (2025). Statistical flaws of the fitness-fatigue sports performance prediction model. *Sci Rep*.

Moxnes, J. F. and Hausken, K. (2008). The dynamics of athletic performance, fitness and fatigue. *Mathematical and Computer Modelling of Dynamical Systems*, 14(6):515–533.